Binary Heart Disease Prediction Using Machine Learning

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***Abstract*—Cardiovascular disease is the leading cause of death worldwide. In this project, we develop a binary classification system that predicts heart disease from 11 routinely collected clinical features. Using a hyperparameter-tuned Random Forest classifier trained on 1,190 patient records, we achieve a test accuracy of 89.1% and an F1 score of 90.2%.**

I. Introduction and Background

A. Motivation

According to the World Health Organization, cardiovascular disease (CVD) causes about 17.9 million deaths worldwide each year, accounting for approximately 32% of total global mortality [1]. In the United States, one person dies from heart disease every 33 seconds. Coronary heart disease is the most prevalent form and causes approximately 375,000 deaths annually. The financial burden of this disease is also substantial; direct medical costs and lost productivity exceed $400 billion each year.

Easily measured risk factors—including high blood pressure, high cholesterol, smoking, diabetes, obesity, and physical inactivity—are known predictors of heart disease. However, the interactions among these factors are complex and often nonlinear, making them difficult for clinicians to fully synthesize during routine examinations. This complexity can lead to missed opportunities for early prevention. Traditional diagnosis often relies on coronary angiography, a costly and invasive procedure. Alternatively, risk assessment may depend on clinical judgment, which can vary across physicians and medical institutions.

B. Machine Learning for Clinical Decision Support

Machine learning thus provides a data-driven approach to identify risk factors simultaneously and to produce objective and reproducible risk assessments. Traditional risk scores are typically based on pre-specified equations derived from the characteristics of the cohort used to develop the model. In contrast, machine learning models adapt to the statistical structure of the training data, allowing them to capture complex relationships and nonlinear interactions among features.

In this domain, ensemble algorithms—particularly Random Forest—have consistently demonstrated strong performance compared with competing approaches. Random Forest constructs an ensemble of decision trees, each trained on a bootstrap sample of the original dataset while considering only a random subset of features at each split. This dual source of randomness reduces variance and decreases susceptibility to overfitting, even when the available dataset is relatively small. Moreover, Random Forest naturally provides feature importance scores, which are especially valuable in clinical contexts because they help clinicians identify the variables that most strongly influence each prediction.

C. Problem Statement

In this project, we develop a **binary classification system** that predicts the presence or absence of heart disease from 11 routinely collected clinical features. Our approach centers on a hyperparameter-tuned Random Forest classifier, chosen for its balance of predictive accuracy, robustness, and interpretability. The complete pipeline encompasses data cleaning and imputation, feature engineering informed by domain knowledge, model training with cross-validated hyperparameter search, and comprehensive evaluation using multiple metrics.

D. Contributions

The key contributions of this project are:

- A robust preprocessing pipeline that handles implicit missing values (zeros coded as missing), engineers clinically motivated features (age groups, heart rate reserve, binary risk indicators), and standardizes continuous variables.
- A systematic model development process using GridSearchCV with stratified 5-fold cross-validation to identify optimal hyperparameters,

evaluated across accuracy, precision, recall, and F1 score.

- Deployment of the trained model as an interactive web application built with Flask, providing real-time predictions with per-patient feature contribution visualizations and identified risk factors.

The remainder of this report is organized as follows: Section II reviews related literature. Section III describes the dataset and presents exploratory data analysis. Section IV details our proposed methodology. Section V presents experimental results. Section VI concludes with discussion and future directions. Section VII provides the project roadmap and team contributions.

II. Literature Review

Our literature review justifies the design choices for this project rather than simply summarizing past studies.

Wilson [4] demonstrated that age, blood pressure, blood cholesterol, smoking, and diabetes are strong determinants of coronary heart disease. This justifies our approach to feature engineering: retaining clinically significant variables as input features rather than selecting variables based purely on statistical correlations.

Hippisley-Cox et al. [5] developed and validated QRISK3 using large real-world cohorts, showing that prediction quality depends not only on algorithm choice but also on how completely and reasonably risk factors are represented. This supports our strategy of combining insights from EDA with feature preprocessing and selection—considering both data patterns and medical relevance before deciding which features enter the final models.

Arnett et al. [6] emphasized in the 2019 ACC/AHA primary prevention guidelines that CVD risk prediction algorithms are intended to enable risk stratification and guide clinical decision-making, and should not be assessed solely by optimizing a single metric. Following this guidance, we report multiple evaluation metrics including precision, recall, and F1 score, rather than relying on accuracy alone.

Recent machine learning reviews further support our modeling approach. Azmi et al. [7] highlighted that nonlinear and ensemble models generally improve cardiovascular risk prediction, but that overfitting, class imbalance, and poor generalization remain persistent challenges. Banerjee et al. [8] made a similar assessment and stressed the importance of interpretability for practical adoption. These findings motivate our plan to first build an interpretable baseline model and then compare stronger nonlinear models under the same validation framework.

III. Dataset Description and Exploratory Data Analysis

A. Data Source

We use the *Heart Disease Dataset* [2] compiled from five well-known cardiac databases: the Cleveland Clinic, the Hungarian Institute of Cardiology, the University Hospital of Zurich, the Long Beach VA Medical Center, and the Statlog project. These databases were originally collected between 1988 and 2019, covering diverse patient populations across different geographic regions. The combined raw dataset contains **1,190 records** with 11 clinical input features and a binary target variable (`target`: 0 = no disease, 1 = disease).

B. Feature Description

The 11 features span five continuous measurements and six categorical or binary indicators, summarized in Table I. These features are routinely collected during standard cardiac examinations—including basic demographics, blood pressure, blood work, and exercise stress test results—making the model applicable in typical clinical settings without requiring specialized or invasive procedures.

TABLE I: Description of the 11 clinical features.

Feature	Type	Description
Age	Cont.	Age in years (28–77)
Sex	Binary	0 = Female, 1 = Male
Chest Pain Type	Cat.	1–4 (typical to asymptomatic)
Resting BP	Cont.	Blood pressure (mmHg)
Cholesterol	Cont.	Serum cholesterol (mg/dl)
Fasting Blood Sugar	Binary	1 if >120 mg/dl
Resting ECG	Cat.	0 = Normal, 1 = ST-T, 2 = LVH
Max Heart Rate	Cont.	Exercise max heart rate
Exercise Angina	Binary	1 if exercise-induced
Oldpeak	Cont.	ST depression (ex. vs. rest)
ST Slope	Cat.	1 = Up, 2 = Flat, 3 = Down

The continuous features capture key cardiovascular health indicators: age reflects cumulative risk exposure, resting blood pressure and cholesterol measure vascular and metabolic health, maximum heart rate reflects cardiac fitness under stress, and oldpeak quantifies exercise-induced ischemia. The categorical and binary features encode clinical observations from ECG readings, symptom assessment, and blood tests.

C. Data Cleaning

During this process, several data quality issues were identified and addressed as follows.

(a) A total of 272 duplicate rows were removed, leaving 918 unique observations. This duplication is understandable because the five source databases contain partially overlapping patient records.

(b) Some features contained zero values that represent implicit missing data rather than valid clinical measurements. Specifically, cholesterol was recorded as zero in 172 cases (a cholesterol level of 0 mg/dl is physiologically impossible), resting blood pressure was zero in one case, and ST slope was zero in one case. These values were replaced with `NaN` to prevent them from biasing subsequent analyses.

(c) Imputation was then performed to fill the resulting missing values. Continuous variables (age, resting blood pressure, cholesterol, maximum heart rate, and oldpeak) were imputed using the column **median**, which is robust to skewed distributions. Categorical variables (sex, chest pain type, fasting blood sugar, resting ECG, exercise angina, and ST slope) were imputed using the column **mode**. Finally, all categorical and binary features were converted to integer format to ensure consistency.

D. Target Distribution

Fig. 1 shows the distribution of the target variable after cleaning. The dataset contains 508 negative cases (no disease) and 682 positive cases (disease), yielding a positive-class ratio of approximately 57%. This moderate imbalance does not necessitate aggressive resampling techniques such as SMOTE, though we use stratified splitting throughout all experiments to preserve class proportions in both training and test sets.

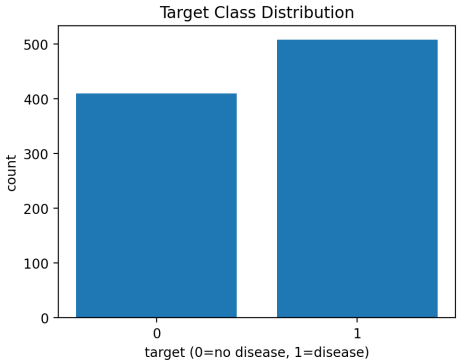


Fig. 1: Target class distribution after cleaning. The dataset is approximately balanced between disease (1) and no-disease (0) classes.

E. Continuous Feature Distributions

Fig. 2 displays histograms of the five continuous features. Age is roughly normally distributed with a mean of 53.7 years ($\sigma = 9.4$), ranging from 28 to 77. Cholesterol shows a right-skewed distribution with a prominent spike near zero due to the implicit missing values (addressed during cleaning); after imputation, the distribution centers around 200–250 mg/dl. Resting blood pressure is concentrated between 120–140 mmHg, consistent with mild hypertension being common in cardiac patients. Oldpeak is heavily right-skewed, with the majority of patients having values near zero and a long tail extending to 6.2, indicating that severe ST depression is relatively rare. Maximum heart rate spans a wide range (60–202 bpm) and is approximately normally distributed, centered around 140 bpm.

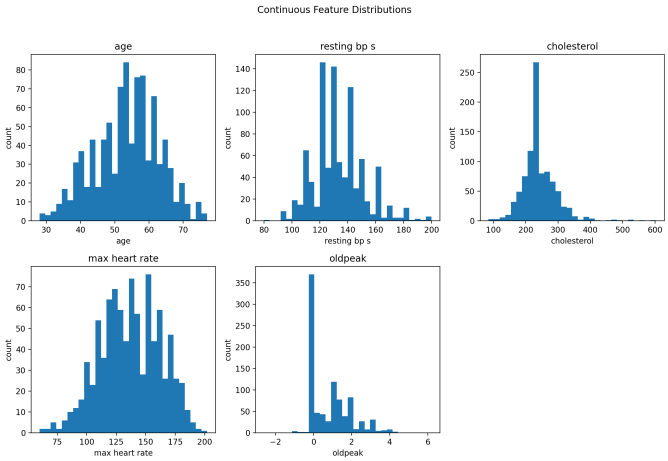


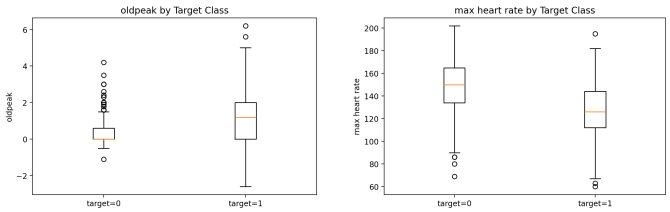
Fig. 2: Distributions of the five continuous features before imputation of zero-as-missing values.

F. Feature–Target Relationships

We examined the relationship between individual features and the target variable to identify the most discriminative predictors.

Fig. 3 shows boxplots for two important continuous features stratified by diagnosis. The median value of *oldpeak* is substantially higher among patients with heart disease (approximately 1.2) compared with those without heart disease (approximately 0.0). This observation is consistent with the clinical understanding that ST depression occurring during or after exercise is an indicator of myocardial ischemia. In contrast, patients without heart disease achieved significantly higher *maximum heart rates* during exercise than patients with heart disease (approximately 158 vs. 130 bpm), reflecting impaired

exercise capacity and a reduced chronotropic response among patients with cardiac disease.



(a) Oldpeak by target class. (b) Max heart rate by target class.

Fig. 3: Box plots of key continuous features stratified by heart disease diagnosis.

Among categorical features, exercise-induced angina emerged as one of the strongest discriminators of disease. As shown in Fig. 4, the disease rate among patients who experience exercise angina is roughly three times higher than among those who do not (approximately 85% vs. 35%), consistent with its role as an important cardiac warning sign. Similarly, chest pain type demonstrated strong discriminative power, with asymptomatic chest pain (type 4) associated with the highest disease prevalence. This finding aligns with clinical evidence that silent ischemia is common among patients with advanced coronary artery disease.

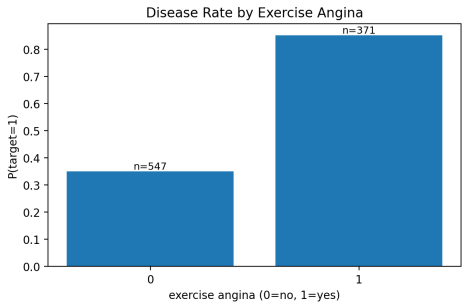


Fig. 4: Disease rate stratified by exercise-induced angina. Patients with angina (1) show substantially higher disease prevalence.

These EDA findings informed our subsequent feature engineering decisions and confirm that the dataset contains clinically meaningful signals suitable for machine learning prediction.

IV. Proposed Methodology

In this research, we used a modeling framework that progresses from linear to advanced ensemble methods to compare the learning process and performance of our heart disease prediction models. We utilized key features analyzed and generated during the feature engineering phase to train each model. By observing the transformation from linear to non-linear relationships between features and heart disease, we explored various ways to maximize generalization ability and improve accuracy on unseen data.

A. Baseline: Logistic Regression

To capture the linear relationship between risk factors and heart disease, we used a Logistic Regression model as our baseline. The largest advantage of Logistic Regression is its transparency and high interpretability—we can observe the coefficients to understand exactly how much an increase in a specific feature affects the probability of heart disease, thus enabling more accurate prevention and early intervention.

The initial stage of Logistic Regression involves aggregating clinical features (such as blood pressure, heart rate, and ST slope) by assigning optimized weights, resulting in a linear combination:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \tag{1}$$

where $x_1, x_2, \ldots, x_n$ represent the independent variables, β_0 is the intercept (bias), and $\beta_1, \ldots, \beta_n$ are the weights learned during training.

The output of the linear combination can be any real number, so it passes through the sigmoid function to map it to a probability value between 0 and 1:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{2}$$

where z is the output of the linear combination. The sigmoid function is an S-shaped curve that maps any real number to the range $(0, 1)$. In this context, the result indicates the probability of a patient having heart disease. If the probability exceeds a decision threshold, the patient is classified as high risk; otherwise, low risk.

B. Non-Linear: Decision Tree

Since Logistic Regression can only handle linearly separable problems, given the potentially complex non-linear relationships between physiological features and heart disease risk, we introduce a Decision Tree model. The Decision Tree is constructed using a *divide-and-conquer* strategy with a *greedy search algorithm*.

The tree selects the optimal feature to split the data at each layer based on a specified criterion. In our model,

the optimal criterion obtained by grid search is Gini Impurity, calculated as:

$$Gini(D) = 1 - \sum_{i=1}^{n} p_i^2 \tag{3}$$

where p_i represents the probability of class i in dataset D, and n is the number of classes ($n = 2$ for binary classification). To identify the best split, the algorithm iterates through every feature and all split points to find the one that minimizes the Gini Impurity. For each split, the Gini index is:

$$Gini_index = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \tag{4}$$

where $|D_1|$ and $|D_2|$ are the sizes of the two subsets. The algorithm chooses the split with the lowest $Gini_index$ as the root node and divides the dataset into D_1 and D_2. This process repeats for each subset, finding the locally optimal split at each step. The tree stops splitting when all samples in a branch share the same label, the sample count falls below $min_samples_split$, or the depth exceeds max_depth. A new sample follows the decision logic at each node until reaching a leaf, where the majority class determines the prediction.

C. Advanced: Random Forest

To address the limitations of a single Decision Tree, which is prone to overfitting, we used a Random Forest model that combines multiple decision trees to reduce variance and improve generalization.

Random Forest reduces variance through bootstrap sampling and random feature selection. Bootstrap sampling randomly selects samples with replacement from the original dataset, repeated B times (where B is the number of estimators). Feature randomization further prevents overfitting: at each node split, the algorithm considers only a random subset of m features (where $m < N$). Through grid search, we determined the optimal subset size to be $\sqrt{N}$.

For a new patient data $\mathbf{x}$, B decision trees each produce a prediction $\hat{y}_b$. The Random Forest determines the final class through majority voting. One key advantage is the ability to compute feature importance via Gini importance. For a feature f, its relative importance V_f is:

$$V_f = \frac{1}{B} \sum_{b=1}^{B} \sum_{n \in nodes(T_b, f)} \Delta Gini(n) \tag{5}$$

where $nodes(T_b, f)$ denotes the nodes in the b-th tree split using feature f, and $\Delta Gini(n)$ represents the Gini impurity reduction at node n. Higher V_f scores indicate features that more strongly influence heart disease risk.

D. Hyperparameter Tuning Strategy

We used `GridSearchCV` with 5-fold stratified cross-validation for all three models. For Logistic Regression, we tuned the regularization strength C with $L2$ penalty (Ridge), which constrains weight magnitudes to prevent overfitting. For the Decision Tree, we tuned maximum depth, minimum samples for splitting, and impurity criterion. For Random Forest, we tuned the number of trees, the number of features per split, and minimum leaf samples. In evaluating results, we focused on *recall* in addition to accuracy and AUC, since in clinical prediction, maximizing recall reduces false negatives and the serious consequences of missing potential heart disease diagnoses.

V. Experimental Results and Evaluation

A. Experimental Setup

To evaluate the effectiveness of the proposed models, the cleaned and engineered dataset was divided into training and testing subsets using an 80-20 stratified split. Stratification ensures that the class distribution between heart disease (positive) and non-disease (negative) cases remains consistent across both subsets.

The training set contains 734 samples, while the testing set contains 184 samples. All model development, including hyperparameter tuning, was performed exclusively on the training set using 5-fold cross-validation. The held-out test set was used only for the final performance evaluation.

B. Performance Metrics

We applied five primary metrics to provide a comprehensive evaluation of the performance of the models. Note that:

- **True Positive (TP)**: The number of positive samples correctly classified as positive.
- **True Negative (TN)**: The number of negative samples correctly classified as negative.
- **False Positive (FP)**: The number of negative samples incorrectly classified as positive.
- **False Negative (FN)**: The number of positive samples incorrectly classified as negative.

These quantities form the basis of the following metrics.

1) **Accuracy**
   Accuracy measures the proportion of correctly classified samples among all observations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

2) **Precision**
Precision measures how many predicted positive samples are actually positive:

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

3) **Recall**
Recall measures the proportion of actual positive samples correctly identified by the model:

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

4) **F1-score**
The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$

5) **AUC-ROC**
The Receiver Operating Characteristic (ROC) curve is a graphical representation used to evaluate the diagnostic ability of a binary classifier. The curve plots the **True Positive Rate (TPR)** against the **False Positive Rate (FPR)** at various classification thresholds.
The True Positive Rate, also known as recall or sensitivity, is defined as

$$TPR = \frac{TP}{TP + FN} \quad (10)$$

The False Positive Rate is defined as

$$FPR = \frac{FP}{FP + TN} \quad (11)$$

By varying the decision threshold of the classifier, different pairs of TPR and FPR values are obtained and this forms the ROC curve.
The Area Under the ROC Curve (AUC) measures the overall ability of the model to distinguish between positive and negative classes. A higher AUC value indicates better classification performance. An AUC value of 0.5 represents random guessing, while a value close to 1.0 indicates excellent discriminative ability.

C. Model Performance Metrics Report

Table II provides a comprehensive horizontal comparison of the models.

TABLE II: Comparative Model Performance Metrics

Model	Acc	Prec	Recall	F1	AUC
Logistic Regression	0.875	0.910	0.860	0.880	0.927
Decision Tree	0.837	0.870	0.830	0.850	0.889
Random Forest	**0.891**	**0.902**	**0.902**	**0.902**	**0.923**

D. Model 1: Logistic Regression

Logistic Regression was implemented as the baseline classification model due to its interpretability and strong performance in many medical prediction tasks. L2 regularization was applied to prevent overfitting. The Logistic Regression model was optimized using a grid search over the regularization parameter C. The best performance was found at $C = 10$.

As summarized in Table II, the model achieved an overall accuracy of 87.5% and a ROC-AUC of 0.927, which indicated strong discrimination between patients with and without heart disease. The model also achieved a precision of 0.91 and a recall of 0.87 for the disease class, which demonstrated reliable identification of high-risk patients.

Though Logistic Regression does not inherently rank features via Gini impurity like ensemble methods, the magnitude and sign of its learned coefficients gives a direct insight into feature influence.

TABLE III: Top Influential Features in Logistic Regression

Feature	Coefficient Value
ST slope_2	2.59
chest pain type_4	1.42
ST slope_3	1.36
fasting blood sugar	1.35
sex	1.35
oldpeak_abnormal	-1.16
exercise angina	0.93
bp_risk	0.92
oldpeak	0.65
chest pain type_3	-0.45

As in Table III, analysis of the model coefficients revealed that *ST slope_2* (2.59) and *chest pain type_4* (1.42) were the strongest positive predictors of heart disease risk. Conversely, features such as *oldpeak_abnormal* (-1.16) show a strong negative correlation. They serve as protective indicators for this model. This linear hierarchy of risk factors provides a transparent baseline for clinical decision-making before considering more complex non-linear ensemble models.

E. Model 2: Decision Tree

Decision Trees were implemented to capture non-linear relationships and provide a hierarchical decision structure for clinical diagnosis. The initial unconstrained tree is shown in Fig. 5. It reached a depth of 11 and achieved a test accuracy of only 76.6%, which are clear signs of overfitting.

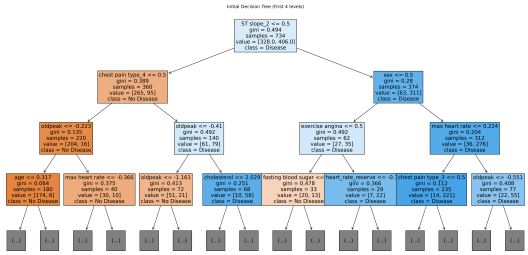


Fig. 5: Initial Decision Tree Before Pruning

To improve generalization, hyperparameter tuning was conducted using GridSearchCV, focusing on *max_depth*, *min_samples_split*, and the split criterion.

The optimized model (Fig. 6) utilized the Gini impurity criterion with a *max_depth* of 3. This pruning significantly improved performance, with a test accuracy of 83.7% and a ROC-AUC of 0.889.

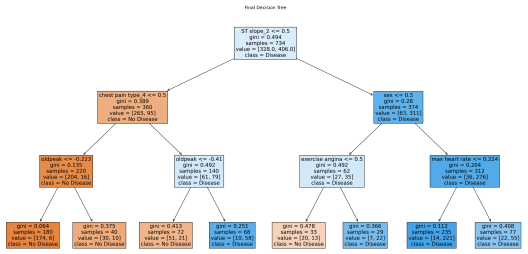


Fig. 6: Pruned Decision Tree Used for Final Evaluation

As detailed in Table II, the model achieved a balanced precision (0.870) and recall (0.830) for the disease class. While the overall metrics are lower than the linear baseline, the pruned tree offers high clinical utility through its transparency. The model's decision logic emphasizes the *ST Slope* and *Chest Pain Type* as the primary root nodes, which aligns with established cardiological diagnostic protocols.

Fig. 7 shows the feature importance ranking from Decision Tree. *ST slope_2* (0.5758) and *chest pain type_4* (0.2014) are the primary drivers of prediction. They collectively contribute nearly 78% of the total model importance. Other variables like max heart rate (0.0289) and exercise angina (0.0200) contribute much less, while the remaining features have negligible impacts on the predictions.

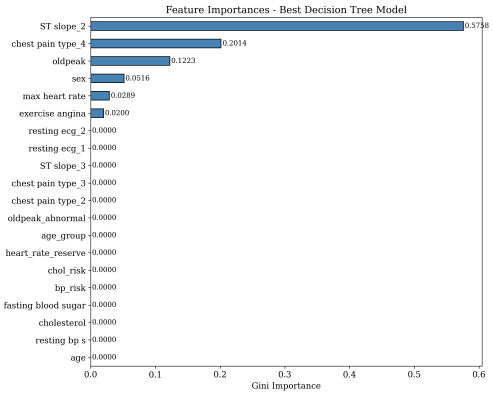


Fig. 7: Feature importance ranking from Decision Tree

F. Model 3: Random Forest

To maximize predictive accuracy and model robustness, a Random Forest classifier was developed as the advanced ensemble model. This approach aggregates multiple decision trees to reduce variance and risk of overfitting.

The model was configured with 200 to 800 estimators and optimized using 5-fold stratified cross-validation. The cross-validation results demonstrated high stability, which maintains a mean F1-score of 0.873 with a low standard deviation of 0.02.

The Random Forest model outperformed all other classifiers and achieved a peak test accuracy of 89.1% and a ROC-AUC of 0.923. Notably, the model achieved perfectly balanced precision and recall scores of 0.902 for the disease class, resulting in an F1-score of 0.902. This balance is critical in medical settings, ensuring that the model is equally effective at identifying true patients (sensitivity) and minimizing false alarms (specificity).

Feature importance analysis (Fig. 8,) revealed that *ST slope_2* (0.1725) and *chest pain type_4* (0.0978) were the most influential predictors of heart disease risk. This indicates the significant diagnostic weight of ST-segment morphology and asymptomatic chest pain in identifying cardiovascular risk. Additionally, the high ranking of the engineered feature *heart_rate_reserve* (0.0833) demonstrates that functional dynamic metrics offer superior predictive power compared to static physiological indicators like resting blood pressure or age.

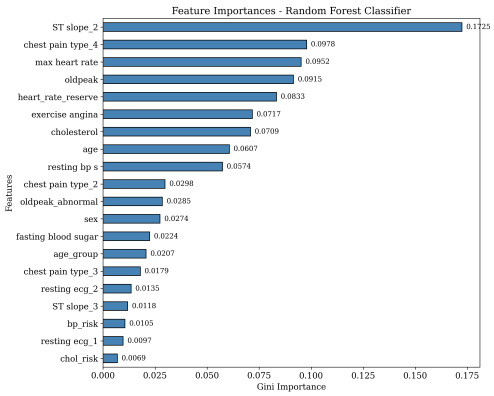


Fig. 8: Feature importance ranking from Random Forest

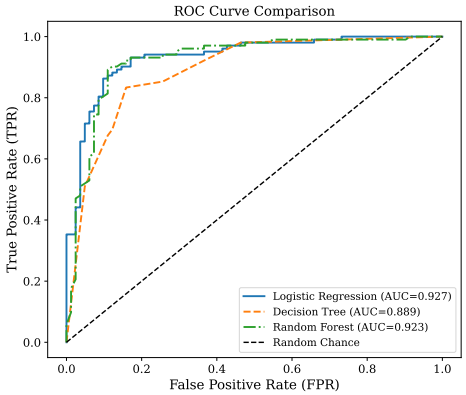


Fig. 9: ROC Curve Comparison Across the Three Models

VI. Comparative Performance Analysis

The performance of the three models generally reveals an evolution in predictive capability as model complexity increases. While all models achieved an accuracy above 83%, their suitability for clinical diagnosis varies depending on their error distributions.

A. Trade-offs Between Precision and Recall

In medical screening, as missing a patient with heart disease can be life-threatening, the cost of a False Negative (FN) is significantly higher than a False Positive (FP).

- Logistic Regression achieved the highest Precision (0.910), making it very reliable when it predicts a positive case. However, its Recall (0.860) was lower than the Random Forest, which suggests it may miss more subtle pathological patterns.
- Decision Tree lagged in both metrics (Precision: 0.870, Recall: 0.830). This indicates that a single tree is often not enough to capture the variance in complex physiological data.
- **Random Forest** achieved an optimal balanced precision and recall scores of both 0.902. This indicates that this ensemble approach is more effective at identifying true disease cases, which reduces the risk of undiagnosed cardiac conditions.

B. ROC and AUC Analysis

The Area Under the Curve (AUC) is a critical metric for evaluating how well the models distinguish between the two classes across all thresholds.

As shown in Fig. 9, **Logistic Regression** yielded the highest AUC (0.927) and slightly outperformed the Random Forest (0.923). This suggests that while the Random Forest is better at the default 0.5 decision threshold, the Logistic Regression model has a slightly superior global separation power.

The Decision Tree's lower AUC (0.889) is likely a result of its discrete nature. Since it makes splits based on binary conditions, its probability estimates do not perform as well as the continuous outputs of Logistic Regression or the averaged probabilities of a Random Forest.

C. Stability and Cross-Validation

To ensure these results were not due to a "lucky" train-test split, we analyzed the 5-Fold Stratified Cross-Validation results. The **Random Forest** demonstrated remarkable stability with an F1 mean of 0.8728 and a low standard deviation of 0.02. This consistency suggests that the model is highly generalizable and would perform reliably on new, unseen patient data from different clinical environments.

D. Feature Sensitivity and Clinical Correlation

Analysis of the top predictors across all three models shows a strong consistency on the importance of the *ST slope* and *Chest Pain Type*.

In the **Random Forest** model, the engineered feature *heart_rate_reserve* ranked as the 6th most important feature (0.056), which is notably higher than raw age or resting blood pressure. This validates our hypothesis that derived physiological markers can provide superior predictive signals for cardiovascular health, instead of just static metrics.

VII. Discussion and Conclusion

A. Discussion

The results suggest that the ensemble model of Random Forest performs better for heart disease prediction because it can capture complex relationships between features. Notably, the consistent importance of the *ST slope* feature across all models aligns with clinical observations that exercise-induced electrocardiogram abnormalities are strong indicators of myocardial ischemia.

The engineered feature *heart_rate_reserve* ranked among the top 10 features in the Random Forest model (Importance: 0.056) and outperformed its constituent raw features like *age*. This suggests that domain-informed feature engineering is as vital as model selection. Moreover, the Logistic Regression model's high AUC (0.927) suggests that the relationship between these physiological markers and heart disease is largely identifiable even with linear methods if the data is properly scaled and preprocessed.

B. Limitations

This study has several limitations. The dataset size is relatively small, which may limit the generalization ability of the models. Moreover, the dataset is collected from specific clinical environments and may not fully represent broader patient populations. Future work should evaluate the models on larger and more diverse medical datasets.

C. Conclusion

In this paper, we developed a robust machine learning pipeline for binary heart disease prediction. By systematically comparing three models, we identified Random Forest as the optimal classifier, which achieved an F1-score of 0.9020. Our results highlight that a combination of ST-segment analysis, chest pain classification, and heart rate reserve provides a powerful diagnostic signal. Future work will focus on integrating these models into real-time clinical decision support systems and exploring deep learning architectures to further improve diagnostic sensitivity in asymptomatic patients.

VIII. GitHub Link, Project Roadmap, and Team Contributions

A. GitHub Repository

The complete source code for this project is publicly available at:

https://github.com/wobuzhidaoqishememingzi/Binary-Heart-Disease-Prediction

The repository contains data preprocessing scripts, exploratory data analysis notebooks, machine learning model implementations, evaluation code, and a Flask-based web application for real-time prediction. Instructions for reproducing all experiments are provided in the repository README.

B. Project Roadmap

The project followed a structured six-week development timeline. Each stage focused on a specific component of the machine learning pipeline, from initial data exploration through final deployment. Table IV summarizes the main milestones.

TABLE IV: Project roadmap and milestones.

Stage	Main Tasks
Week 1 (Late Jan)	Define project objective, select dataset, perform initial data inspection and cleaning, begin exploratory data analysis.
Week 2 (Early Feb)	Conduct detailed EDA, identify data quality issues, perform preprocessing, begin feature selection.
Week 3 (Mid Feb)	Finalize features and preprocessing pipeline, implement baseline models, establish training setup.
Week 4 (Late Feb)	Train multiple ML models, evaluate with cross-validation and classification metrics.
Week 5 (Early Mar)	Analyze results, generate performance plots, interpret model behavior.
Week 6 (Mid Mar)	Integrate components, develop web interface, finalize report and presentation.

C. Team Contributions

The project was completed collaboratively by five team members. Each member focused on specific stages of the pipeline while also contributing to discussions, debugging, and integration. Table V summarizes the responsibilities.

TABLE V: Team member responsibilities.

Member	Contributions
Yiran Wang	Dataset acquisition, data cleaning, and exploratory data analysis including feature distributions and target-variable relationships.
Tingwei Zhang	Feature selection, data preprocessing (normalization, encoding), and frontend implementation of the web-based demo system.
Xinyi Li	Baseline model (Logistic Regression) development, multi-model performance evaluation and comparison, and related report writing.
Yuebi Li	Non-linear model (Decision Tree) implementation and methodology report writing.
Victor Zhang	Random Forest model development, backend logic for the web prediction system (preprocessing integration, model loading, API), and report compilation.

Throughout the project, the team maintained regular communication to coordinate progress, review experimental results, and prepare the final deliverables.

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